JONATHAN BERNARD

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Portfolio: https://github.com/jonathanvikash-prog

SUMMARY

Mechatronics Engineering undergraduate (BEng Hons, ongoing) with a Higher Diploma and hands-on experience in robotics, embedded systems, AI-driven automation, and OT cybersecurity. Skilled in ROS 2, ESP32, computer vision, industrial network monitoring, asset discovery, and ICS/SCADA security. Interested in robotics, industrial automation, and OT security.

EXPERIENCE

Operational Technology (OT) Security Engineer — Intern | **Jun 2026 – Present**

Innovar (Private) Limited, Sri Lanka

- Gained hands-on experience in OT/ICS cybersecurity, supporting security monitoring and protection of industrial environments.
- Worked with ICS/SCADA security concepts, industrial network monitoring, and OT asset discovery.
- Assisted with vulnerability assessment and security analysis of OT environments.
- Developed practical knowledge of OT network security, asset visibility, and cybersecurity best practices.
- Applied industry cybersecurity frameworks and standards relevant to industrial control systems and critical infrastructure.
- Gained exposure to OT security technologies including Nozomi Networks and Fortinet.

Technologies & Skills: OT/ICS Security · ICS/SCADA · Nozomi Networks · Fortinet · Network Security · Asset Discovery · Vulnerability Assessment · Industrial Network Monitoring

EDUCATION

BEng. (Hons) Mechatronics and Autonomous Systems-(Ongoing) | **07/2026 - 07/2027**

- Liverpool John Moores University

Higher Diploma in Mechatronics and Autonomous-Systems | **03/2024 - 03/2026**

- Liverpool John Moores University

G.C.E Advanced Level (A/L) | **08/2020 - 01/2023**

- St Benedict's College

PROJECTS

AI-Based Vegetable Sorting System - HD Final Project

- Developed an intelligent sorting system for carrots and brinjals using YOLOv8s for real-time detection.
- Controlled NEMA 17 motors for the conveyor belt and rotary wheel for one-by-one vegetable dispensing.
- Used ESP32 to send signals to servo motors via UDP based on YOLO outputs for automated sorting.

Skills: YOLOv8s, ESP32, Servo Control, NEMA 17 Motors, UDP Communication, Real-time Robotics Automation

TECHNICAL SKILLS

- **Programming Languages:** C++, Python, Embedded C
- **Robotics, Simulation & Computer Vision:** ROS 2 HUMBLE – robot simulation (Gazebo), visualization (RViz), URDF modeling, and SLAM, TensorFlow, YOLO object detection algorithm, Edge Impulse
- **Industrial Automation:** Pure and Electro pneumatic systems, PLC-based automation
- **Design, Tools & Platforms:** SolidWorks, Automation Studio, Proteus, Ubuntu, Arduino IDE, mikroC PRO for PIC
- **Certifications:** Fortinet NSE 1, 2 & 3 Certified in Cybersecurity — Fortinet, 2026,OT Security 7.6 Architect – Lessons — Fortinet, 2026 MATLAB onramp, Cambridge English Level 1